Information Dynamics and Applications
https://www.acadlore.com/journals/IDA

ACADLORE
Global Scholarship, Curated with Rigor

Kernel-Based Approach to Multitask Problems: A Review of Concepts, Methods and Applications

Nengem Stephen Mkegh[1*], Donald Torsabo[2*]

[1] Department of Mathematics and Statistics, Taraba State University, PMB 1167 Jalingo, Nigeria
[2] Department of Fisheries and Aquaculture, Joseph Saawuan Tarka University, PMB 2373 Makurdi, Nigeria

[*] Correspondence: Nengem Stephen Mkegh (nengem.stephen@tsuniversity.edu.ng);
Donald Torsabo (donald.torsabo@uam.edu.ng)

Received: 04-29-2026 **Revised:** 06-06-2026 **Accepted:** 06-20-2026

Citation: N. S. Mkegh and D. Torsabo, "Kernel-based approach to multitask problems: A review of concepts, methods and applications," *Inf. Dyn. Appl.*, vol. 5, no. 2, pp. 72–83, 2026. https://doi.org/10.56578/ida050201.

© 2026 by the author(s). Licensee Acadlore Publishing Services Limited, Hong Kong. This article can be downloaded for free, and reused and quoted with a citation of the original published version, under the CC BY 4.0 license.

Abstract: Multitask learning (MTL) is a machine learning paradigm in which several related tasks are learned simultaneously to improve generalization performance. Kernel-based methods provide a mathematically rigorous and flexible framework for MTL, especially when training data are limited or uncertainty estimation is important. This study examined the research landscape of MTL using the Scopus database. The findings reveal that MTL has experienced remarkable growth in recent years, with over 93% of publications produced between 2016 and 2026, demonstrating its increasing relevance in modern scientific and technological research. The subject-area restriction further confirmed the highly interdisciplinary nature of the field, particularly across computer vision, medical imaging, predictive analytics, and artificial intelligence applications. Despite the rapid expansion of deep learning-based multitask approaches, the analysis identified only a very limited number of studies specifically focused on kernel-based MTL, indicating a significant research gap within the literature. This scarcity suggests that kernel-based methods remain largely underexplored despite their strong mathematical foundations, interpretability, and effectiveness in nonlinear modeling. The study therefore concludes that kernel-based MTL presents substantial opportunities for future theoretical development and practical applications, making it a promising direction for advancing MTL research.

Keywords: Multitask learning; Kernel-based methods; Support vector machines; Gaussian processes; Multiple kernel learning; Reproducing kernel Hilbert space; Machine learning

1 Introduction

Kernel-based methods for multitask problems have emerged as a highly important research area in modern machine learning due to their ability to learn multiple related tasks at once while effectively sharing information between them. Traditional single-task learning handles each task independently, often missing useful connections that may exist across tasks. In contrast, multitask learning (MTL) enhances prediction accuracy and generalization by leveraging shared structures and dependencies among tasks [1]. This collaborative learning approach is especially beneficial when data is limited, noisy, or high-dimensional. Kernel methods such as support vector machines (SVMs), Gaussian processes (GPs), and multiple kernel learning (MKL) offer a robust mathematical framework for capturing nonlinear patterns through implicit mappings into high-dimensional spaces. As a result, combining kernel techniques with MTL has led to powerful models that can tackle complex real-world challenges in areas like computer vision, bioinformatics, recommendation systems, medical diagnosis, speech recognition, cybersecurity, and natural language processing [2].

The main advantage of kernel-based MTL is its ability to represent task relationships through shared kernels, task covariance matrices, regularization techniques, and reproducing kernel Hilbert spaces (RKHS). Rather than learning each task in isolation, kernel-based MTL models jointly optimize multiple related objectives, enabling useful knowledge to be transferred between tasks. This sharing mechanism often yields greater robustness, less overfitting, and superior predictive performance compared to independent single-task methods [3]. Additionally, kernel methods offer theoretical benefits such as convex optimization properties, strong statistical learning guarantees, and flexibility in handling nonlinear and structured data. Unlike many deep learning models that demand massive datasets and heavy computational resources, kernel-based methods tend to work well on small to medium-sized datasets while preserving interpretability and stability. Recent research also indicates that multitask kernel frameworks can effectively incorporate diverse data sources and prior domain knowledge into the learning process [4].

In recent years, there have been major advancements in developing advanced kernel-based MTL techniques. Researchers have introduced sophisticated regularization-based models, adaptive kernel selection strategies, logic-constrained multitask frameworks, and hybrid systems that merge kernel methods with deep neural networks. A foundational research article improved the numerical stability and accuracy of kernel-based approximation methods, which are fundamental to many modern kernel learning algorithms [5]. These advances have influenced MTL by enabling more efficient representations of shared feature spaces and operator-valued kernels used to model relationships among multiple tasks. Consequently, the paper provides essential mathematical and computational tools that underpin the development of scalable kernel-based MTL methods. The research provides a foundation for other emerging concepts such as learning to learn, transfer learning, and learning with privileged information, which have greatly broadened the applicability of kernel-based MTL systems. Moreover, recent work has explored MTL under logical constraints, where prior logical relationships among tasks are embedded directly into the optimization process to boost performance in sparse, high-dimensional settings. These developments have reinforced the importance of kernel-based MTL in intelligent systems and data-driven decision-making [6].

Despite these significant advances, several challenges continue to limit the full potential of kernel-based MTL approaches. A key concern is negative transfer, where unrelated tasks harm the performance of other tasks during joint learning. Other issues include computational demands, scalability to large datasets, kernel selection, optimization difficulties, and accurately capturing task relationships [7]. Furthermore, integrating kernel-based MTL methods with modern AI paradigms such as deep learning, continual learning, and federated learning remains an open area of research. Therefore, there is a growing need for comprehensive reviews that cover recent theoretical developments, methodologies, practical applications, and unresolved challenges in kernel-based MTL. This review paper aims to provide a systematic overview of kernel-based approaches to MTL, focusing on recent advances, mathematical foundations, applications, and future research directions in the field.

2 Fundamentals of Kernel Methods

Kernel methods rely on mapping input data into higher-dimensional feature spaces. Fasshauer is a prominent figure in kernel-based approximation theory, especially known for his work on radial basis functions (RBFs), reproducing kernels, meshfree methods, and kernel interpolation. While his research does not focus exclusively on MTL, many of the theoretical concepts he developed have greatly influenced modern kernel-based MTL frameworks. His contributions offer the mathematical foundation that enables multiple related tasks to share information within a common RKHS [7].

Central to Fasshauer's approach is the idea that kernels function as similarity measures, allowing data to be embedded into high-dimensional spaces where complex nonlinear patterns become easier to model. In multitask settings, several related learning objectives are solved together instead of separately. Fasshauer's kernel philosophy supports this by creating shared functional spaces where different tasks can interact through common kernels or correlated basis functions. The framework relies heavily on positive definite-kernels RBFs like Gaussian, Multiquadric, Inverse Multiquadric, and Matérn kernels. These kernels define smooth approximation spaces suitable for interpolation, regression, classification, and coupled learning systems [8].

A key contribution from Fasshauer is the unification of RBF approximation and moving least squares (MLS) approximation within a single reproducing kernel framework. In his influential work on dual bases and discrete reproducing kernels, he showed that RBF approximation can be viewed as a constrained optimization problem in Hilbert spaces. This perspective is relevant to MTL because multitask systems also depend on coupled optimization principles, where multiple tasks are regularized together. The shared kernel structure allows related tasks to transfer knowledge through common representations while still maintaining task-specific characteristics [5].

Mathematically, Fasshauer's framework begins with a kernel function,

$$K(x, y) = \varphi(\|x - y\|) \tag{1}$$

where, φ is a RBF and $(\|x - y\|)$ measures the distance between two points. The approximation of a function is then expressed as

$$f(x) = \sum_{i=1}^{N} \lambda_i K(x, x_i) \tag{2}$$

where, the coefficients λ_i are determined from training data. In multitask problems; this Eq.(2) is extended by introducing task indices so that several related functions are learned simultaneously:

$$f_t(x) = \sum_{i=1}^{N} \sum_{s=1}^{T} \lambda_{is} K((x, t), (x_i, s)) \tag{3}$$

where, t and s denote tasks. The kernel now captures both input similarity and task similarity. This shared-kerne construction is foundational in modern kernel-based MTL [7].

Another vivid aspect of Fasshauer's work is his emphasis on RKHSs. He showed that kernels are not merely computational tools but are deeply connected to Sobolev spaces, Green's functions, and differential operators. His Green-kernel approach explains how kernels naturally encode smoothness properties and differential constraints. In MTL, this becomes extremely valuable because tasks often share structural properties such as smoothness, continuity, or correlated dynamics. By embedding tasks into a common RKHS, multitask models can achieve improved generalization and stability, especially when data are sparse [7].

A key strength of Fasshauer's kernel-based philosophy is regularization [7]. MTL requires balancing shared knowledge against task individuality. Fasshauer's RKHS framework achieves this naturally through norm minimization and smoothness penalties. The optimization problem typically takes the form,

$$\min_{f \in \mathcal{H}} \sum_{i=1}^{N} L\left(y_i, f\left(x_i\right)\right) + \lambda \|f\|_{\mathcal{H}}^2 \tag{4}$$

where, L is a loss function and $\lambda\|f\|_{\mathcal{H}}^2$ controls smoothness and complexity. In multitask settings, coupled regularization terms encourage related tasks to remain close in the shared Hilbert space. This idea later became central to regularization-based MTL methods discussed in modern surveys of kernel-based MTL [7].

In summary, Fasshauer's contribution to kernel-based MTL can be characterized as a mathematically rigorous framework grounded in RBFs, reproducing kernels, Hilbert space theory, and meshfree approximation. His methods prioritize smoothness, adaptability, shared representation, and regularized optimization all of which are key components in today's MTL systems. Although modern MTL has progressed toward deep kernel and hybrid architectures, much of its theoretical underpinnings remain strongly linked to Fasshauer's foundational work on kernel approximation and reproducing kernel spaces.

Currently, kernel-based approaches to MTL have been greatly advanced by several other leading researchers who laid the theoretical and computational groundwork for contemporary multitask systems. The article demonstrated that dynamic graph neural networks can effectively model nonlinear dependencies and task interactions in multi-task prediction by learning adaptive graph representations between related outputs [9]. Although the proposed approach does not employ kernel functions or RKHS, it addresses the same fundamental objective as kernel-based MTL, leveraging shared information across correlated tasks to improve predictive accuracy. The study therefore highlights the recent shift from traditional kernel-based approaches, which rely on predefined covariance or similarity kernels, toward graph-based deep neural architectures capable of automatically learning complex inter-task relationships from data [9].

Evgeniou et al. [10] introduced one of the earliest regularization-based multitask frameworks in their influential paper "Learning Multiple Tasks with Kernel Methods." They proposed that each task's function can be broken down into a shared global component and a task-specific part,

$$f_t(x) = f_0(x) + g_t(x) \tag{5}$$

thereby allowing related tasks to transfer knowledge while maintaining individual characteristics. His work demonstrated that joint learning improves generalization especially when training data are limited [10].

Micchelli made major contributions through the development of vector-valued and operator-valued kernels. He extended scalar kernels into matrix-valued forms,

$$K(x, y) \in R^{T \times T} \tag{6}$$

where, kernel entries describe relationships among tasks. This work provided rigorous representation theorems and mathematical foundations for multitask RKHS theory, which later became central in structured prediction and multi-output learning [11].

Another leading scholar is Massimiliano, who focused on multitask feature learning and sparse regularization. Together with collaborators, he developed convex optimization techniques that encourage tasks to share informative features through mixed-norm regularization methods. His framework improved feature selection and prediction accuracy in applications such as bioinformatics and computer vision [11].

Argyriou further advanced sparse MTL by introducing shared latent feature representations [12]. His matrix factorization framework expressed multitask models as,

$$W = UV \tag{7}$$

where, U represents shared latent features and V contains task-specific coefficients. His methods allowed efficient learning of common structures among related tasks while reducing model complexity.

In probabilistic MTL, Edwin Bonilla introduced influential GP multitask models. He proposed covariance kernels of the form,

$$K\left((x,t),(x',t')\right) = K_x\left(x,x'\right) K_t\left(t,t'\right) \tag{8}$$

where, one kernel measures input similarity while the other captures task similarity. This tensor-product kernel framework became highly influential in multitask GPs, transfer learning, and uncertainty-aware prediction systems [13].

Together with Fasshauer, these scholars transformed kernel-based MTL into a mathematically rigorous field grounded in RKHS theory, convex optimization, GPs, sparse learning, and operator-valued kernels. Their contributions continue to influence modern developments in transfer learning, deep kernel methods, scientific machine learning, and artificial intelligence.

3 Concept of Multitask Learning

Over the past few years, the field of MTL has seen rapid theoretical and practical progress, driven largely by advances in deep learning, foundation models, and multimodal AI systems [14]. MTL is a machine learning approach where several related tasks are learned together using shared representations and joint optimization. Recent efforts have concentrated on improving task generalization, minimizing negative transfer, refining parameter-sharing mechanisms, and embedding MTL into large-scale foundation models. Numerous leading researchers have shaped the current understanding of MTL through surveys, theoretical models, optimization techniques, and practical applications [15, 16].

One of the most impactful recent works is a comprehensive survey by Yu and colleagues, titled "Unleashing the Power of Multi-Task Learning," which offers a broad modern overview of MTL's evolution from traditional methods to foundation-model-based approaches. Their work categorizes MTL into several types: regularization-based learning, relationship learning, feature propagation, optimization strategies, and pre-training frameworks [17]. They argue that today's MTL has moved beyond fixed-task architectures toward flexible, prompt-driven systems capable of handling diverse data types and zero-shot generalization. A key strength of their contribution is linking classical MTL with modern large language models and multimodal systems. However, the survey remains largely theoretical and taxonomical, with limited empirical comparisons across domains. Additionally, it leaves open questions about scalability and interpretability in very large multitask systems. Nonetheless, the paper has become a foundational resource in current MTL research by unifying decades of scattered literature into a coherent structure [17].

Another highly influential contribution is the survey by Zhang et al. [18], which continues to underpin many current MTL methods. Although their foundational survey predates the last two years, recent studies frequently build on his classification of feature learning, task clustering, low-rank modeling, and task relationship learning. Contemporary researchers still rely heavily on their mathematical formulations of shared task structures and task covariance learning. The lasting relevance of his work highlights its robustness and theoretical clarity. Still, critics note that the framework was developed primarily for classical machine learning and does not fully accommodate transformer-based architectures or foundation models that now dominate AI. Despite this limitation, the analytical framework remains central to much of the recent MTL optimization literature [18].

A third key contribution comes from Crawshaw [19], whose work on deep neural network MTL is still widely cited in recent MTL developments. Crawshaw systematically categorized deep MTL into architectural methods, optimization methods, and task relationship learning. Recent researchers continue to extend his analysis of hard and soft parameter sharing, adversarial task balancing, and gradient conflict mitigation. The strength of Crawshaw's work lies in its clarity and accessibility, serving as a bridge between classical statistical MTL and modern deep architectures. However, recent progress in foundation models and multimodal systems has exposed some limitations, particularly the framework's insufficient treatment of transformer scaling laws and prompt-based learning. Nevertheless, many studies from 2024 to 2025 still use Crawshaw's structural classification as a baseline theoretical framework [19].

In the area of kernel-based MTL, kernel-based approximation methods and RKHS provide an important theoretical foundation for model development [7]. More broadly, recent advances in MTL with deep neural networks, including task relationships, architectures, and optimization strategies, have been systematically reviewed [19]. The survey emphasized that kernel approaches remain mathematically rigorous and interpretable compared to deep neural architectures. Kernel methods are especially effective for small-data scientific applications where explainability is crucial. However, the review also acknowledges serious drawbacks: kernel methods often struggle with scalability on high dimensional datasets and are computationally expensive for large multitask settings. Moreover, kernel-based approaches are increasingly being overshadowed by transformer-based multitask systems in industrial AI. Despite these limitations, the work shows that kernel methods still offer strong theoretical guarantees and remain valuable in healthcare, signal processing, and scientific computing [20].

Another recent contribution is the work of Koryakovskaya et al. [21], who proposed a semi-supervised cross-domain MTL framework for psychological profile analysis from text data. The study addresses the practical challenge of incomplete annotations by enabling joint learning from two separately annotated datasets, one for emotion recognition and another for personality trait assessment, without requiring additional labeling. To achieve this, the authors combined pre-trained language encoders, cross-attention mechanisms, and confidence-based pseudo-labeling within a hybrid supervised and semi-supervised learning framework. A major contribution of the study is its demonstration that effective knowledge transfer can be achieved across related tasks despite the absence of jointly annotated data, making the approach valuable for personalization systems, human AI interaction, and computational psychometrics. However, the proposed framework is specifically designed for psychological profile analysis from text and may require adaptation before it can be applied to other multitask domains such as reinforcement learning or general scientific computing. Overall, the study represents a significant advancement in semi-supervised MTL under realistic annotation scarcity [21].

Recent domain-specific surveys also show that MTL is becoming deeply integrated into specialized fields. For example, studies on recommender systems and minimally invasive surgical vision emphasize gradient balancing, mixture-of-experts architectures, and conflict-aware optimization. Collectively, these works indicate that modern MTL research is shifting from simple parameter sharing toward adaptive and dynamic task coordination. Researchers are now prioritizing optimization stability, task compatibility estimation, and scalable multimodal learning over simply improving predictive accuracy [22].

Critically, despite remarkable progress, the literature reveals persistent unresolved challenges in MTL. Negative transfer remains a major issue when unrelated tasks interfere destructively during optimization. Another concern is the lack of universally accepted metrics for measuring task relatedness and transfer efficiency. Furthermore, the computational demands of large multitask foundation models raise concerns about energy consumption, scalability, and interpretability. Researchers also continue to debate whether multitask systems truly generalize better than highly specialized single-task models in all scenarios [23].

Overall, recent years have seen substantial theoretical and practical growth in MTL research. Scholars such as Yu, Zhang, Crawshaw, Fasshauer, and many others have advanced the field through theoretical unification, optimization strategies, kernel methodologies, and practical domain applications. Their contributions reveal that MTL is evolving from a specialized machine learning technique into a central paradigm for future artificial intelligence systems capable of handling heterogeneous, multimodal, and large-scale tasks simultaneously.

4 Kernel-Based Approaches to Multitask Problems

Kernel-based approaches to MTL have become one of the most mathematically rigorous and practically influential directions in modern machine learning. MTL itself refers to the simultaneous learning of multiple related tasks with the objective of improving predictive performance through shared representations and transfer of knowledge among tasks. Unlike single-task learning, where each problem is solved independently, MTL exploits latent relationships among tasks to enhance generalization, particularly in situations involving limited or noisy data [24]. Over the past five years, kernel-based MTL has evolved rapidly due to the increasing demand for robust, interpretable, and theoretically grounded artificial intelligence systems.

The origin of kernel-based MTL can be traced to the emergence of SVMs and RKHS theory in statistical learning. Early multitask frameworks relied on regularization theory and vector-valued kernels to jointly model several tasks within a unified functional space. These methods provided elegant mathematical structures for learning shared task representations while preserving task individuality. Classical scalar kernels of the form,

$$K\left(x, x'\right) \tag{9}$$

were generalized into matrix-valued kernels as,

$$K\left(x, x'\right) \cdot \in R^{T \times T} \tag{10}$$

where, T denotes the number of tasks and the matrix structure encodes inter-task relationships. This formulation enabled multitask problems to be represented within vector-valued RKHS and became foundational to later developments in kernel-based MTL systems.

Between 2021 and 2026, research on kernel-based MTL experienced a major transition from traditional convex optimization frameworks toward hybrid architectures integrating deep learning, operator theory, tensor analysis, and probabilistic learning. One of the most influential developments during this period was the renewed interest in vector-valued kernels and reinforcement learning systems [25], in their comprehensive Scopus-indexed survey published in "Applied Soft Computing", classified modern multitask methods into feature-based, combination-based, and regularization-based approaches. Their work demonstrated that recent kernel-based MTL systems no longer rely solely on fixed similarity structures but instead incorporate adaptive and data-driven mechanisms for task sharing.

Feature-based kernel methods for MTL became increasingly important because of their ability to construct common latent representations across heterogeneous datasets. Modern approaches utilize nonlinear kernel embeddings to map data into high-dimensional feature spaces where related tasks become easier to learn jointly. These methods are especially useful in applications such as medical diagnosis, climate prediction, bioinformatics, and natural language processing where tasks often share hidden structural patterns. Deep feature extraction combined with kernel mappings has significantly improved the representational capacity of multitask systems [26].

Another major development in the last five years has been the growth of MKL in multitask environments. Rather than relying on a single kernel function, contemporary models combine several kernels adaptively to capture different characteristics of the data. Dynamic kernel weighting, sparse kernel combinations, and task-adaptive fusion mechanisms have allowed multitask systems to selectively share information among related tasks while minimizing negative transfer from unrelated tasks. This flexibility has improved predictive accuracy in complex learning environments involving multimodal and high-dimensional data [27].

Regularization-based MTL remains one of the strongest theoretical pillars of kernel approaches. Most recent models are formulated as optimization problems of the form,

$$\min_{f_1,\dots f_T} \sum_{t=1}^{T} L_t\left(f_t\right) + \lambda\Omega\left(f_1,\dots,f_T\right) \tag{11}$$

where, L_t represents task-specific loss functions and Ω denotes a multitask regularization operator controlling the level of task coupling. Modern regularization methods increasingly incorporate graph Laplacians, spectral penalties, low-rank structures, and operator constraints to improve task coordination. Such formulations have proven highly effective in domains where task relationships evolve dynamically over time [25, 28].

A particularly important advancement since 2021 is the emergence of deep kernel MTL. In these systems, deep neural networks are integrated with kernel methods to combine the representational strength of deep learning with the theoretical robustness of RKHS frameworks. The resulting models often take the following form [18, 29].

$$K\left(x_i, x_j\right) = k\left(\phi_\theta\left(x_i\right), \phi_\theta\left(x_j\right)\right) \tag{12}$$

where, ϕ_θ denotes a neural representation learned through deep architectures and $k(\cdot,\cdot)$ is a kernel function. Deep kernel MTL systems have shown strong performance in image analysis, healthcare prediction, financial forecasting, and intelligent control systems. Their ability to provide uncertainty estimation alongside nonlinear representation learning has become particularly valuable in safety-critical applications.

Recent years have also witnessed increased attention toward tensor and manifold kernel methods. Tensor RKHS frameworks allow MTL over multidimensional relational structures and are especially useful in multimodal and spatiotemporal systems. Related kernel-theoretic studies, including work on the Hilbert–Schmidt singular value decomposition and iterated Brownian bridge kernels [5], provide useful mathematical foundations for spectral analysis in structured kernel spaces. Modern manifold kernel methods further incorporate geometric learning principles to capture nonlinear task relationships embedded in complex data spaces.

Despite these advances, several challenges continue to affect kernel-based MTL. Scalability remains one of the most serious problems because kernel methods typically require operations involving large kernel matrices with computational complexity proportional to $O(n^3)$. Although recent approximation methods such as Nyström sampling, sparse kernels, and random feature mappings have reduced computational burdens, scalability remains a central research challenge in large-scale multitask environments [29].

Another persistent issue is negative transfer, which occurs when unrelated tasks degrade each other's performance. Contemporary research therefore focuses heavily on adaptive task grouping, attention-guided task relations, and dynamic weighting strategies that allow multitask systems to identify beneficial task interactions automatically. Interpretability also remains an important concern, especially in hybrid deep-kernel systems where the transparency traditionally associated with kernel methods may be partially lost [30].

At present, kernel-based MTL occupies a highly interdisciplinary position linking machine learning, functional analysis, optimization theory, Bayesian inference, and operator theory. Current systems increasingly integrate graph learning, attention mechanisms, probabilistic inference, and spectral analysis into kernel-based MTL frameworks. The field remains highly relevant because kernel methods provide strong mathematical guarantees, robust generalization on limited datasets, and reliable uncertainty quantification that many purely deep learning systems struggle to achieve [29, 31].

Future directions in kernel-based MTL are expected to focus on quantum kernels, physics-informed learning, federated multitask systems, and operator-based artificial intelligence. Quantum kernel methods may provide richer feature embeddings and computational advantages for extremely high-dimensional multitask problems [32]. Physics-informed kernels are also expected to become important in scientific machine learning, renewable energy modeling, and climate systems where physical laws can be embedded directly into learning architectures. Similarly, federated kernel MTL is emerging as a promising solution for privacy-preserving collaborative learning in healthcare, smart cities, and Internet of Things environments.

5 Methodology for the Quantitative Analysis of Kernel-Based Approaches to Multitask Learning Research

Multi-task learning (MTL) has attracted considerable interest in the machine learning community because it enhances model performance by transferring knowledge among several related tasks. With the rising popularity of MTL, it is important to comprehend its development trajectory, methodological strategies, and diverse range of applications [14, 32]. This part provides a quantitative review of published research on MTL and kernel-based methods using "Scopus," a highly reputable database for scientific literature. The aim is to examine the progression of MTL research from 2016 to 2026 and to identify its main application areas. This search was carried out in May 2026. Relevant studies on multi-task learning were selected using specific inclusion and exclusion criteria. The analysis was carried out using the Scopus database. The search strategy involved terms such as "multi-task problems" or "multitask problems," "multi-task learning" or "multitask learning," "kernel-based approach" or "kernel based approach," and "kernel-based method" or "kernel based method," applied to titles, abstracts, and keyword fields. The search was limited to publications released between 2016 and 2026 across all disciplines, and then further restricted to key subject areas such as mathematics, computer science, engineering, physics, and medicine. To maintain high standards of quality and relevance, only journal articles, review papers, and book chapters were included in the final set.

6 Discussion and Analysis of Searched Results

The bibliometric search results obtained on May 26, 2026 from the Scopus database on the topic "multitask learning" provide significant insight into the development, research intensity, and specialization trends within the field. The general unrestricted search yielded a total of 11,557 publications, while restricting the publication period to 2016–2026 reduced the number slightly to 10,831 publications. Further restriction to specified subject areas to mathematics, computer science, engineering, physics, and medicine produced 4,743 publications. Most importantly, when the search was narrowed specifically to "kernel-based approaches to multitask learning", only 9 publications were identified. This final observation is particularly important because it reveals a substantial research gap despite the overall rapid growth of MTL research. Consequently, the reported number of nine publications refers to the studies that satisfied the review criteria and should not be interpreted as the total number of papers related to broader areas such as multi-output learning, operator-valued kernels, or multitask GPs, which often extend beyond the intended focus of this review.

The first major implication of the results is the extraordinary expansion of MTL research within the last decade. Out of the total 11,557 publications, 10,831 appeared between 2016 and 2026, representing approximately 93.7% of the entire literature. This indicates that MTL is predominantly a contemporary research area whose growth coincides strongly with the global advancement of artificial intelligence, machine learning, and deep learning technologies. The sharp concentration of publications within this recent period suggests that MTL has become one of the most active methodological paradigms in modern computational science.

This rapid growth can largely be attributed to the increasing limitations of traditional single-task learning systems. In many real-world applications, related tasks often share common structures and information. MTL addresses this challenge by enabling simultaneous learning across multiple related tasks, thereby improving generalization, reducing overfitting, and enhancing predictive performance. As a result, researchers in areas such as computer vision, speech recognition, natural language processing, healthcare analytics, robotics, finance, and bioinformatics increasingly adopted MTL frameworks.

The reduction from 10,831 publications to 4,743 publications after restricting the search to only mathematics, computer science, engineering, physics, and medicine further demonstrates the interdisciplinary spread of MTL research. This indicates that a large portion of the literature exists outside the core computational disciplines commonly associated with artificial intelligence. In essence, MTL has evolved beyond a purely theoretical machine learning concept into a broadly applicable analytical methodology used across scientific and engineering domains.

However, the most analytically significant finding emerges from the final restriction concerning "kernel-based approach to multitask learning", where only 9 publications were identified. Compared to the thousands of publications on general MTL, this number is extremely small and immediately suggests that kernel-based MTL remains an underexplored niche within the broader field. Statistically, 9 publications constitute less than 0.2% of the 4,743 subject-restricted publications and an even smaller fraction of the total MTL literature. This sharp disparity indicates that although MTL itself has become highly popular, kernel-based formulations have not received equivalent research attention.

This finding is particularly noteworthy because kernel methods historically played a foundational role in machine learning before the widespread dominance of deep neural networks. Kernel-based learning approaches, especially those associated with RKHS, SVMs, and RBF methods, are known for their strong theoretical guarantees, mathematical interpretability, and effectiveness in handling nonlinear structures. Their limited representation in MTL literature therefore reveals a potentially important research opportunity rather than a lack of relevance.

One possible explanation for the low number of kernel-based MTL publications is the overwhelming dominance of deep learning architectures after 2015. Modern research trends heavily favor deep neural networks due to their scalability and success in large-data environments. Consequently, many researchers shifted attention away from classical kernel methods toward transformer architectures, convolutional neural networks, graph neural networks, and reinforcement learning frameworks. While these approaches achieved remarkable practical success, they often sacrifice interpretability and theoretical transparency, areas where kernel methods remain particularly strong.

Another explanation may be the computational complexity traditionally associated with kernel methods, especially when applied to large-scale multitask problems. Kernel matrices can become computationally expensive for massive datasets, making deep learning appear more practical in industrial applications. Nevertheless, advances in sparse kernels, low-rank approximations, adaptive kernels, and distributed optimization techniques now provide opportunities to overcome these limitations. Therefore, the small number of identified publications may indicate that the area is still emerging rather than obsolete.

From a research perspective, the scarcity of kernel-based MTL studies creates a strong justification for further investigation. A field with only 9 identified publications offers substantial potential for originality, innovation, and scholarly contribution. This is particularly important for researchers seeking to develop mathematically rigorous and theoretically interpretable MTL frameworks. Kernel-based approaches possess several advantages that remain highly relevant, including robustness in small-data scenarios, flexibility in nonlinear mapping, and compatibility with optimization theory and functional analysis.

Furthermore, the gap suggests the possibility of integrating modern deep learning concepts with classical kernel methodologies to create hybrid multitask systems. Such integration could combine the representational strength of deep architectures with the theoretical stability and generalization properties of kernel methods. Emerging areas such as multiple kernel MTL, adaptive radial basis kernels, multitask GPs, and kernelized transfer learning therefore represent promising future research directions.

The bibliometric findings also indicate that kernel-based MTL has not yet reached research maturity. Unlike the general MTL literature, which already exhibits characteristics of a rapidly expanding and highly competitive field, kernel-based MTL remains relatively undeveloped. This lack of saturation is advantageous for new researchers because it allows room for theoretical contributions, algorithmic development, comparative analyses, and application-oriented innovations without facing excessive redundancy in existing literature.

In conclusion, the Scopus search results reveal that MTL is a rapidly growing and highly influential research area, with more than 93% of its publications appearing between 2016 and 2026. The reduction from 11,557 publications to 4,743 after subject-area restriction confirms the broad interdisciplinary application of the field. Most significantly, the discovery of only 9 publications specifically addressing kernel-based approaches to MTL highlights a major research gap within the literature. This scarcity suggests that kernel-based MTL remains an underexplored yet potentially valuable area of study. Given the strong mathematical foundations, interpretability, and nonlinear modeling capabilities of kernel methods, future investigations in this direction could contribute substantially to the advancement of MTL theory and applications.

7 Interpretation and Discussion of Results

The graphs present a comprehensive bibliometric visualization of the research landscape in MTL and reveal the major thematic directions, relationships, and developmental trends within the field. Figure 1 demonstrates a highly interconnected network structure in which the size of nodes, the density of links, and the clustering patterns collectively indicate the prominence and relationships among research themes. The dominance of keywords such as "image" and "segmentation" clearly shows that MTL research is heavily concentrated in computer vision and image processing applications. Their central positions and strong interconnections suggest that image segmentation, classification, object detection, and feature extraction constitute the core of current MTL studies. The presence of associated terms such as "feature map," "convolution," "detector," and "super resolution" further confirms the extensive reliance on deep learning architectures, particularly convolutional neural networks, in solving multitask problems.

The visualization also reveals the multidisciplinary nature of MTL through the existence of several interconnected thematic clusters. One major cluster is associated with medical imaging and healthcare applications, containing terms related to diagnosis, MRI, ultrasound imaging, lesions, tomography, and biomarkers. This indicates that MTL has become increasingly important in automated medical diagnosis and clinical decision-making systems. The close association between diagnostic and segmentation-related terms implies that segmentation techniques are often integrated as supporting tasks in intelligent medical analysis systems. Such integration improves efficiency, accuracy, and predictive capability in healthcare applications.

Another noticeable thematic direction is the emergence of forecasting and predictive analytics applications. Keywords related to forecasting, load prediction, simulation, RMSE, and optimization indicate that MTL is being widely employed in engineering systems, energy demand forecasting, and predictive modeling. This demonstrates that the methodology has expanded beyond image-based problems into broader scientific and industrial applications. The inclusion of terms associated with optimization and intelligent systems further suggests that MTL is increasingly connected with reinforcement learning, autonomous decision-making, and adaptive computational systems.

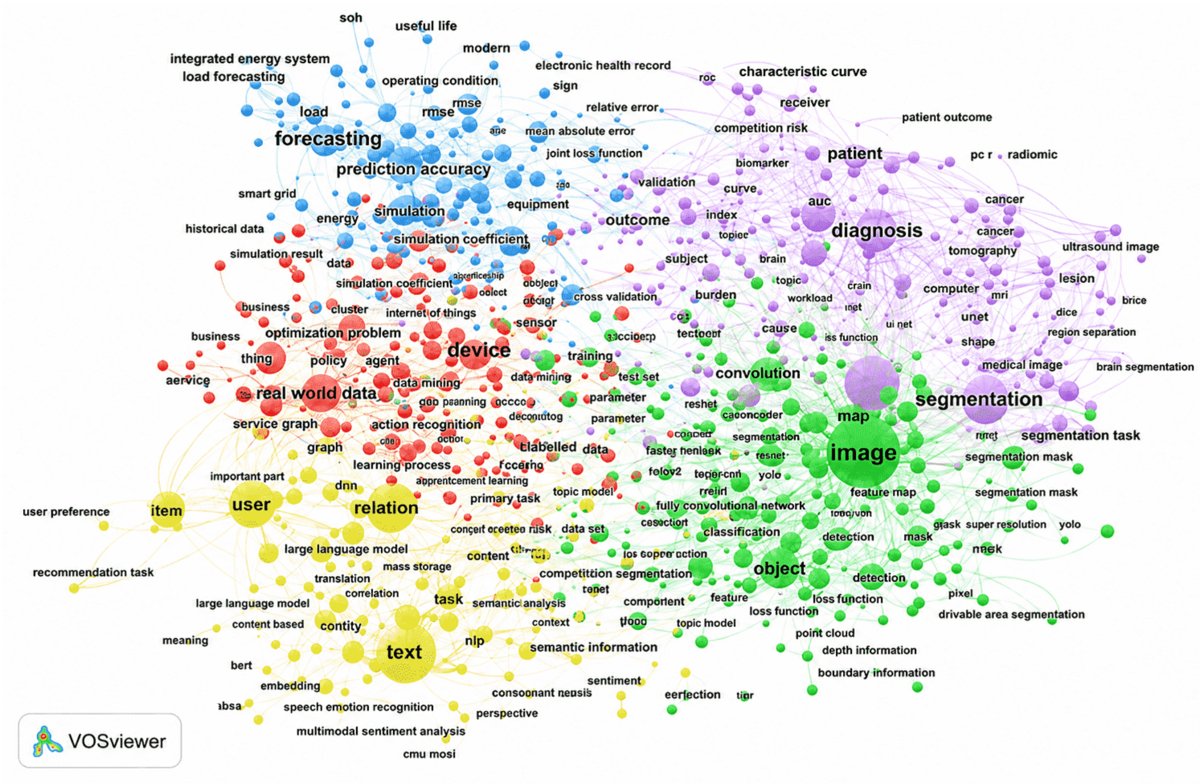

Figure 1. Multitask learning (MTL) performance network visualization

The graph additionally highlights the growing influence of natural language processing and multimodal learning within multitask research. The appearance of terms such as "text," "semantic information," "transformers," "large language model," and "multimodal sentiment analysis" reflects recent advancements driven by modern artificial intelligence architectures. This suggests that MTL is progressively being integrated into language understanding systems and multimodal frameworks capable of processing images, text, and other forms of data simultaneously. The emergence of these themes demonstrates the transition of MTL from traditional single-domain applications toward more generalized artificial intelligence systems.

An important observation from the network is the relatively limited presence of kernel-related approaches despite the enormous volume of MTL publications. The small representation of kernel-based terminology indicates that kernel methods constitute only a minor portion of the broader MTL literature. This finding strongly supports the argument that contemporary research is overwhelmingly dominated by deep learning approaches. Nevertheless, the continued existence of kernel-related terms within the network shows that kernel methods still possess theoretical and methodological relevance, particularly in areas requiring mathematical rigor, interpretability, and optimization efficiency.

Figure 2, which represents an overlay visualization, provides additional insight into the temporal evolution of research themes within MTL. The color variations indicate shifts in research focus over time, with darker colors representing older themes and brighter colors indicating more recent developments. The keyword "kernel" appears in darker coloration, suggesting that kernel-based approaches were more prominent in earlier stages, but there is a shift in research attention toward newer deep learning methodologies. In contrast, newer and brighter themes are associated with broader problem-solving frameworks and emerging artificial intelligence technologies. This temporal transition reflects the global movement in machine learning research from traditional statistical and kernel-based techniques toward data-driven deep neural architectures.

Despite this apparent decline in prominence, the persistence of kernel-related connections indicates that kernel methods remain foundational within the theoretical framework of MTL. Their continued relevance suggests the existence of important research opportunities, particularly in developing hybrid systems that combine the theoretical strengths of kernel methods with the representational power of deep learning models. Such a gap presents a significant avenue for future investigation, especially in applications where interpretability, generalization, and computational stability are essential.

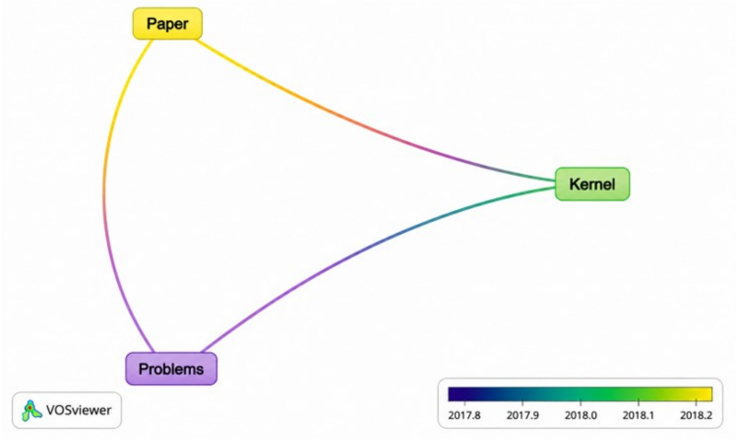

Figure 2. Kernel based approach to multitask learning (MTL) visualization

8 Conclusions

Overall, the graph visualizations demonstrate that MTL is a rapidly evolving and highly interdisciplinary research field with broad applicability, characterized by strong dominance in computer vision, medical imaging, predictive analytics, and modern artificial intelligence systems. The findings further reveal that while deep learning currently drives the majority of research activity, kernel-based MTL remains relatively less investigated compared with deep-learning-based MTL frameworks, with considerable potential for theoretical advancement and practical applications.

Author Contributions

Conceptualization, N. S. M.; methodology, N. S. M.; validation, D. T.; formal analysis, N. S. M.; investigation, N. S. M.; resources, D. T.; data curation, N. S. M.; writing—original draft preparation, N. S. M.; writing—review and editing, D. T.; visualization, D. T. All authors have read and agreed to the published version of the manuscript.

Data Availability

No new primary datasets were generated during this study. The bibliographic records analyzed in this review were retrieved from the Scopus database. The Scopus search records and bibliometric analysis files are available from the corresponding author upon reasonable request.

Acknowledgements

The researchers hereby acknowledged the department of Mathematics and Statistics, Taraba State University Jalingo, for the enabling environment for the research.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors used AI assisted tools to support the editing of portions of this manuscript, including improving language quality, sentence structure, and organization. The AI tools were not used for data collection, data analysis, image generation, statistical analysis, or the formulation of scientific conclusions. Every AI-generated suggestion was critically evaluated, edited, and validated by the authors. The authors accept full responsibility for the accuracy, authenticity, and integrity of all content in this article.

References

[1] M. M. Abdelsamie, S. S. Azab, and H. A. Hefny, "Deep multi-task learning: A review of concepts, methods, and cross-domain applications," *Int. J. Data Sci. Anal.*, vol. 21, no. 1, p. 77, 2026. https://doi.org/10.1007/s41060-025-00892-y

[2] E. Bernard, Y. Jiao, E. Scornet, V. Stoven, T. Walter, and J. P. Vert, "Kernel multitask regression for toxicogenetics," *Mol. Inform.*, vol. 36, no. 10, p. 1700053, 2017. https://doi.org/10.1002/minf.201700053

[3] J. Zhou, L. Gao, C. Lu, and X. Yao, "Collaborative optimization of manufacturing service allocation via multi-task transfer learning evolutionary approach," *J. Intell. Manuf.*, vol. 36, no. 3, pp. 1761–1779, 2025. https://doi.org/10.1007/s10845-024-02339-w

[4] M. Zhou, X. Huang, H. Liu, and D. Zheng, "Hospitalization patient forecasting based on multi-task deep learning," *Int. J. Appl. Math. Comput. Sci.*, vol. 33, no. 1, pp. 151–162, 2023. https://doi.org/10.34768/amcs-2023-0012

[5] R. Cavoretto, G. E. Fasshauer, and M. McCourt, "An introduction to the Hilbert–Schmidt SVD using iterated Brownian bridge kernels," *Numer. Algorithms*, vol. 68, no. 2, pp. 393–422, 2015. https://doi.org/10.1007/s11075-014-9850-z

[6] Z. Zhang, Y. Song, Y. Tan, S. Yan, B. Zhang, and Y. Zhuang, "Segmentation-assisted prostate cancer grading with multitask collaborative learning," *Pattern Recognit. Lett.*, vol. 183, pp. 42–48, 2024. https://doi.org/10.1016/j.patrec.2024.04.023

[7] G. E. Fasshauer and M. J. McCourt, *Kernel-Based Approximation Methods Using MATLAB*. World Scientific, 2015. https://doi.org/10.1142/9335

[8] G. E. Fasshauer and Q. Ye, "A kernel-based collocation method for elliptic partial differential equations with random coefficients," in *Monte Carlo and Quasi-Monte Carlo Methods 2012*. Springer Berlin Heidelberg, 2013, vol. 65, pp. 331–347. https://doi.org/10.1007/978-3-642-41095-6_14

[9] K. Fu and Z. Ren, "Multi-task prediction of wind speed and time-varying wind shear coefficient using dynamic graph interactive neural network," *Inf. Fusion*, vol. 125, p. 103478, 2026. https://doi.org/10.1016/j.inffus.2025.103478

[10] T. Evgeniou, C. A. Micchelli, and M. Pontil, "Learning multiple tasks with kernel methods," *J. Mach. Learn. Res.*, vol. 6, pp. 615–637, 2005. https://doi.org/10.5555/1046920.1088693

[11] C. A. Micchelli and M. Pontil, "On learning vector-valued functions," *Neural Comput.*, vol. 17, no. 1, pp. 177–204, 2005. https://doi.org/10.1162/0899766052530802

[12] A. Argyriou, R. Hauser, C. A. Micchelli, and M. Pontil, "A DC-programming algorithm for kernel selection," in *Proceedings of the 23rd International Conference on Machine Learning*, Pittsburgh, Pennsylvania, USA, 2006, pp. 41–48. https://doi.org/10.1145/1143844.1143850

[13] E. V. Bonilla, K. M. A. Chai, and C. K. I. Williams, "Multi-task Gaussian process prediction," in *Advances in Neural Information Processing Systems 20*, Vancouver, British Columbia, Canada, 2008, pp. 153–160.

[14] P. Cao, X. Liu, H. Liu, J. Yang, D. Zhao, M. Huang, and O. Zaiane, "Generalized fused group lasso regularized multi-task feature learning for predicting cognitive outcomes in Alzheimer's disease," *Comput. Methods Programs Biomed.*, vol. 162, pp. 19–45, 2018. https://doi.org/10.1016/j.cmpb.2018.04.028

[15] G. Dai, C. Guo, T. Zhou, J. Chen, Y. Shao, R. Xu, K. Mao, and X. Qian, "Precise bone age assessment via multi-task feature disentanglement and quadruplet ranking," *Neurocomputing*, vol. 679, p. 133161, 2026. https://doi.org/10.1016/j.neucom.2026.133161

[16] C. Ruiz, C. M. Alaíz, and J. R. Dorronsoro, "A survey on kernel-based multi-task learning," *Neurocomputing*, vol. 577, p. 127255, 2024. https://doi.org/10.1016/j.neucom.2024.127255

[17] J. Yu, Y. Dai, X. Liu, J. Huang, Y. Shen, K. Zhang, R. Zhou, E. Adhikarla, W. Ye, Y. Liu *et al.*, "Unleashing the power of multi-task learning: A comprehensive survey spanning traditional, deep, and pretrained foundation model eras," *arXiv Preprint arXiv:2404.18961*, pp. 1–116, 2024. https://doi.org/10.48550/arXiv.2404.18961

[18] Y. Zhang, L. Yan, and G. Li, "Reinforcement learning-based resource allocation for multi-task multi-codebook semantic communications," *Phys. Commun.*, vol. 75, p. 102982, 2026. https://doi.org/10.1016/j.phycom.2025.102982

[19] M. Crawshaw, "Multi-task learning with deep neural networks: A survey," *arXiv preprint arXiv:2009.09796*, pp. 1–43, 2020. https://doi.org/10.48550/arXiv.2009.09796

[20] C. Li, M. Georgiopoulos, and G. C. Anagnostopoulos, "A unifying framework for typical multitask multiple kernel learning problems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 7, pp. 1287–1297, 2014. https://doi.org/10.1109/TNNLS.2013.2291772

[21] D. O. Koryakovskaya, A. A. Axyonov, E. V. Ryumina, and D. A. Ryumin, "Multi-task human's psychological profile analysis based on text data using semi-supervised learning," *Sci. Tech. J. Inf. Technol. Mech. Opt.*, vol. 26, no. 2, pp. 337–348, 2026. https://doi.org/10.17586/2226-1494-2026-26-2-337-348

[22] Y. Liu, J. Li, M. Gong, Y. Zhang, and Z. Tang, "Towards evolution-based multi-level collaborative multi-task sparse learning," *Swarm Evol. Comput.*, vol. 105, p. 102385, 2026. https://doi.org/10.1016/j.swevo.2026.102385

[23] M. Jiang, J. Hu, J. Zheng, J. Wang, and X. Ye, "Comprehensive multitask ensemble segmentation and clinical interpretation of pancreatic and peripancreatic anatomy with radiomics and deep learning features," *Int. J. Imaging Syst. Technol.*, vol. 36, no. 1, p. e70270, 2026. https://doi.org/10.1002/ima.70270

[24] Z. Huang, Q. Yang, and Y. Gao, "A multi-task learning network with boundary regression for automatic

ultrasound image segmentation," *Biomed. Signal Process. Control*, vol. 113, p. 108828, 2026. https://doi.org/10.1016/j.bspc.2025.108828

[25] F. Zhao and Y. Jia, "A multi-agent reinforcement learning with multi-task learning and graph edge attention networks for dynamic flexible job shop scheduling," *Appl. Soft Comput.*, vol. 187, p. 114277, 2026. https://doi.org/10.1016/j.asoc.2025.114277

[26] W. Li, "On resolving non-preemptivity in multitask scheduling: An optimal algorithm in deterministic and stochastic worlds," *IEEE Trans. Netw.*, vol. 34, pp. 2733–2745, 2026. https://doi.org/10.1109/TON.2025.3647752

[27] C. Li and L. Ma, "Early weak anomaly detection and uncertainty quantification of equipment based on multi-task and multi-domain temporal memory autoencoder," *Eng. Appl. Artif. Intell.*, vol. 162, p. 112735, 2025. https://doi.org/10.1016/j.engappai.2025.112735

[28] C. Song, H. Yang, J. Cai, P. Yang, H. Bao, K. Xu, and X.-B. Meng, "Multi-energy load forecasting via hierarchical multi-task learning and spatiotemporal attention," *Appl. Energy*, vol. 373, p. 123788, 2024. https://doi.org/10.1016/j.apenergy.2024.123788

[29] K. Tang, S. Chen, T. Guo, Y. Ma, and A. J. Khattak, "An adaptive deep multi-task learning approach for citywide travel time collaborative estimation," *Expert Syst. Appl.*, vol. 244, p. 123009, 2024. https://doi.org/10.1016/j.eswa.2023.123009

[30] H. Xu, X. Li, Z. Qian, H. Cheng, and W. Ding, "Multi-task deep learning-empowered in-situ monitoring methodology for visible and invisible defects in laser melt injection: Ceramic reinforced metal matrix composite," *J. Intell. Manuf.*, vol. 37, no. 7, pp. 2767–2786, 2025. https://doi.org/10.1007/s10845-025-02663-9

[31] C. Wang, F. Shi, Y. Di, and H. Liang, "Multi-task learning enhanced deep neural networks for solving delay differential equations," *Commun. Nonlinear Sci. Numer. Simul.*, vol. 153, p. 109528, 2026. https://doi.org/10.1016/j.cnsns.2025.109528

[32] J. Zhu, Y. Liao, Y. Chen, F. Gao, R. Li, G. Yang, X. Lai, and J. Chen, "Multimodal MRI-based glioma segmentation and MGMT promoter methylation status prediction using multitask learning architecture," *Int. J. Imaging Syst. Technol.*, vol. 35, no. 5, p. e70173, 2025. https://doi.org/10.1002/ima.70173